HARDWARE MANUAL / VERSION 0.1

AUSELIA

ADS7057 test board

How a perfboard signal becomes SPI samples, one step at a time.

The topology, the datasheet facts that matter, the routing of the board, and what each part is and where it sits.

Version 0.1 / October 15, 2026 / Nicolás Villegas

How to read this manual

Read the text first. Then look at the figure. The figure shows the same thing with numbers.

01

One panel, one step

A page holds four steps in a 2 × 2 grid. Each panel has a title, a text and a figure.

02

Example tag

A figure with this tag is a worked example. Its numbers come from the netlist, the board file or the simulation.

03

Source

The KiCad project in hw/, the simulations in sim/, the vendor datasheets and one real JLCPCB quote.

04

Estimate

A number marked ESTIMATE is not measured. Read it as a size, not as a promise.

05

Names

R16 is a reference designator. It is the same name in the schematic, the BOM and the silkscreen file.

06

Order

Part 0 gives the basics. Steps 1 to 54 follow the power, the signal, the ADC, SPI, the wake-up, the board and the bring-up.

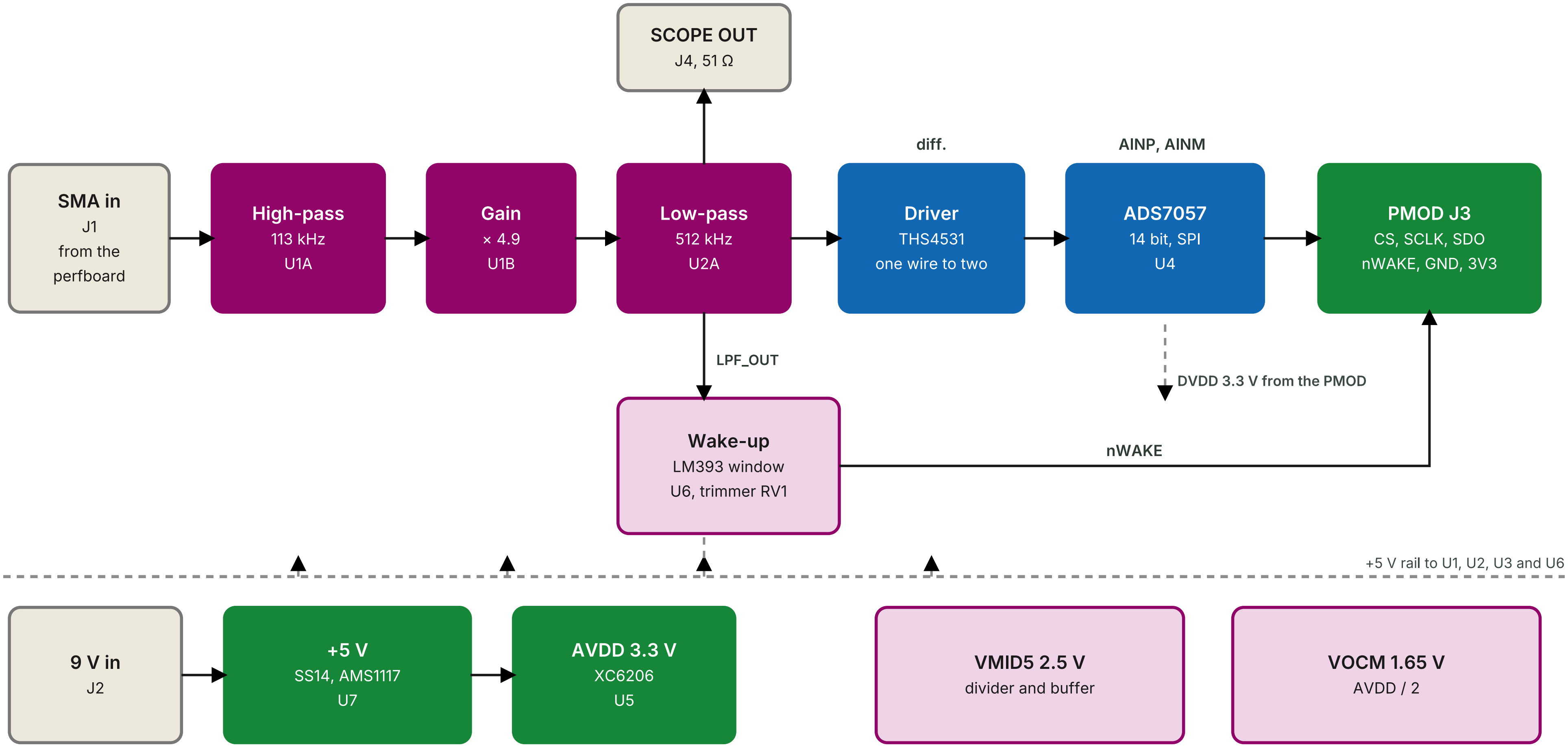
Colors

- 🟪 signal chain
- 🟦 driver and ADC
- 🟩 power
- ⬜ control, data of the example
- 🟨 the item of this step

THE BOARD IN ONE VIEW

The board in one view

the signal goes left to right, the power runs along the bottom



SCOPE OUT
J4, 51 Ω

SMA in
J1
from the perfboard

High-pass
113 kHz
U1A

Gain
× 4.9
U1B

Low-pass
512 kHz
U2A

diff.

Driver
THS4531
one wire to two

AINP, AINM

ADS7057
14 bit, SPI
U4

PMOD J3
CS, SCLK, SDO
nWAKE, GND, 3V3

LPF_OUT

DVDD 3.3 V from the PMOD

Wake-up
LM393 window
U6, trimmer RV1

nWAKE

+5 V rail to U1, U2, U3 and U6

9 V in
J2

+5 V
SS14, AMS1117
U7

AVDD 3.3 V
XC6206
U5

VMID5 2.5 V
divider and buffer

VOCM 1.65 V
AVDD / 2

THE BASICS

part 0

What the signal is, why the mid-rails, what an ADC measures and why the filter comes first.

POWER

steps 1 to 6

9 V to 5 V to 3.3 V, then the 2.5 V and 1.65 V mid-rails that the signal rides on.

SIGNAL AND ADC

steps 7 to 24

High-pass, gain, low-pass, then the driver and the ADC with its input filter.

SPI AND WAKE-UP

steps 25 to 35

The four SPI wires, the frame and its power-up rule, and the window comparator.

THE BOARD

steps 36 to 48

Layers, floor plan, every part and where it sits, routing, files for the factory and the cost.

BRING-UP

steps 49 to 54

Power-on checks, the first signal, the expected response and what is still open.

PART 0

The basics

0.1 TO 0.6

What the board is for, what the signal looks like, why there are two mid-rails, what an ADC measures, why the filter comes first and how to read a PCB. No board details yet.

0.1

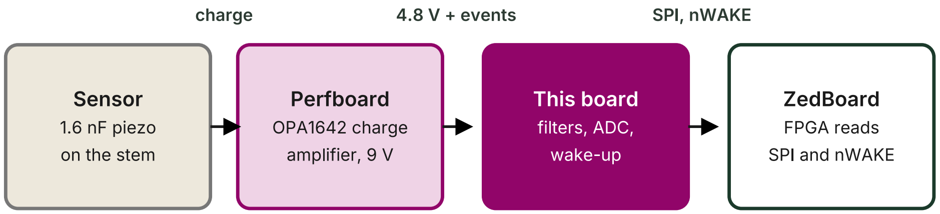
THE BASICS

What the board is for

A piezoelectric sensor on a plant stem hears the clicks of **acoustic emission** (AE). The perfboard charge amplifier turns the charge of the sensor into a voltage.

This board takes that voltage, filters it and measures it with a 14-bit ADC. It hands the numbers to the ZedBoard over SPI. It also raises a wake-up line when a click is strong.

CHAIN


charge
4.8 V + events
SPI, nWAKE
Sensor
1.6 nF piezo on the stem
Perfboard
OPA1642 charge amplifier, 9 V
This board
filters, ADC, wake-up
ZedBoard
FPGA reads SPI and nWAKE

the gray box is the part that this manual describes

0.2

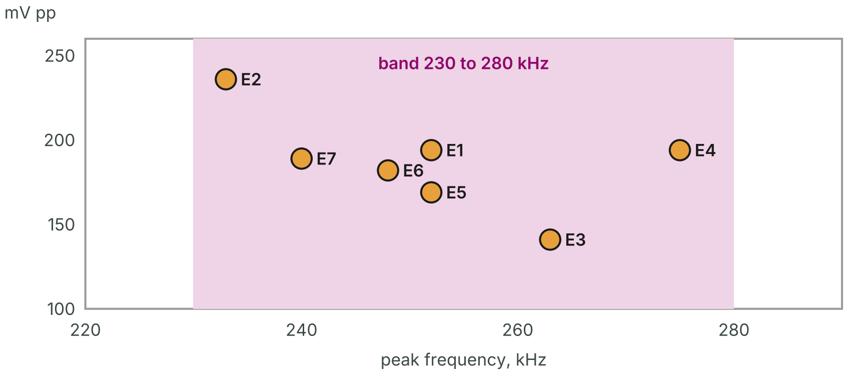
THE BASICS

What the signal looks like

The perfboard output rests near 4.8 V. An AE event rides on it as a burst between 233 and 275 kHz that lasts 26 to 83 µs. The seven recorded events are 141 to 236 mV peak to peak.

Between events, five of the eight baseline records show 3.7 to 4.6 mV rms of noise in the band. That noise sets the smallest event that this board can see.

7 RECORDED EVENTS



0.3

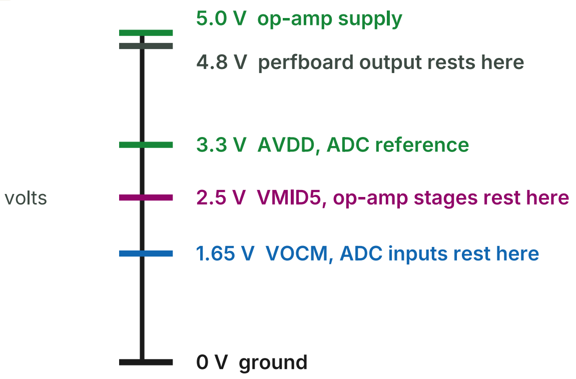
THE BASICS

Why 2.5 V and 1.65 V

This board has no negative supply. A signal that swings below 0 V would be cut off. So each stage rests at the middle of its own supply, and the signal swings around that level.

The op-amps run from 0 to 5 V, so they rest at **2.5 V**. The ADC runs from 0 to 3.3 V, so its inputs rest at **1.65 V**.

VOLTAGE LADDER


volts
5.0 V op-amp supply
4.8 V perfboard output rests here
3.3 V AVDD, ADC reference
2.5 V VMID5, op-amp stages rest here
1.65 V VOCM, ADC inputs rest here
0 V ground

0.4

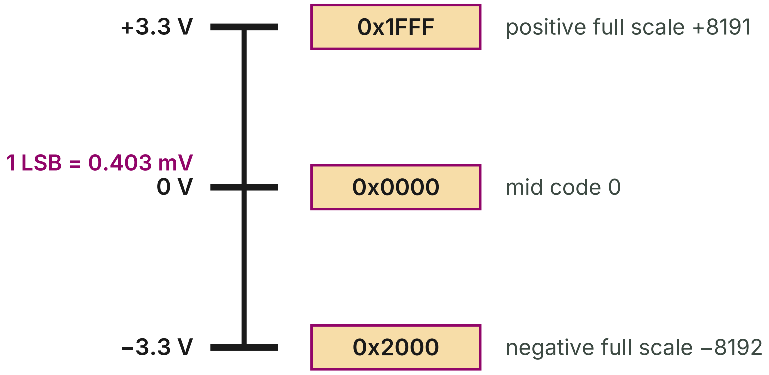
THE BASICS

What an ADC measures

The ADS7057 measures the **difference** between two pins, AINP and AINM. It reports the difference as a 14-bit signed number in two's complement. One step, called 1 LSB, is 0.403 mV.

The full-scale range is plus and minus the supply AVDD, so ±3.3 V. The code 0x0000 means no difference.

CODE LADDER


+3.3 V
0x1FFF
positive full scale +8191
1 LSB = 0.403 mV
0 V
0x0000
mid code 0
−3.3 V
0x2000
negative full scale −8192

0.5

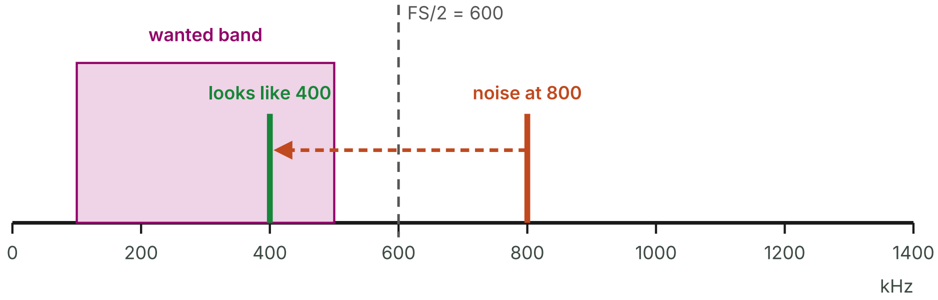
THE BASICS

Why filter before sampling

A sampled signal cannot tell a frequency f from FS − f. At 1.2 MSPS a tone at 800 kHz looks exactly like a tone at 400 kHz, inside the band of interest.

So a low-pass filter must remove what lies above half the sample rate before the ADC sees it. The sample rate here is about 1.2 MSPS with a 25 MHz SPI clock (ESTIMATE).

ALIASING



NOTES

0.6

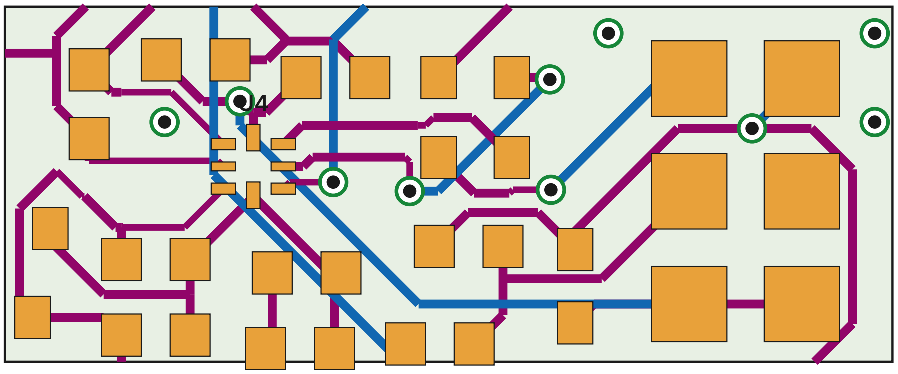
THE BASICS

How to read a PCB

The board has two copper layers. A **pad** holds a part. A **track** joins pads. A **via** is a plated hole that joins the two layers. A **pour** fills free space with copper, here ground.

The **silkscreen** prints the names. The figure is a 20 × 8 mm cut of this board around the ADC.

REAL BOARD, 6 ×



amber: pads purple: top tracks blue: bottom tracks green ring: vias pale green: ground pour

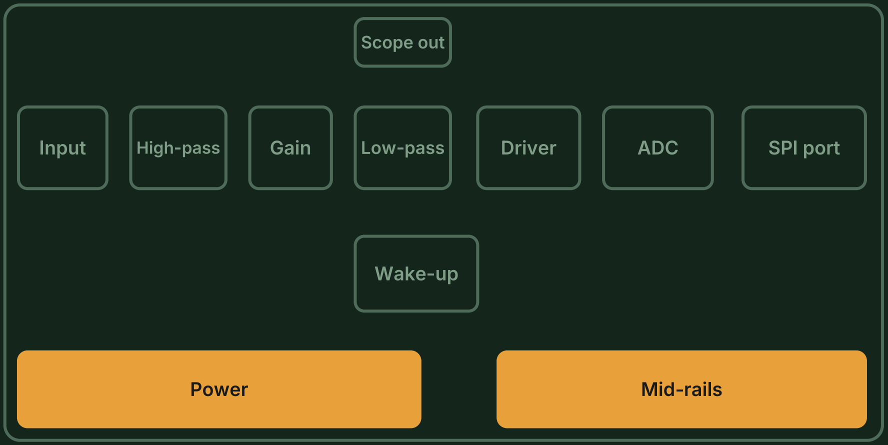
NOTES

SECTION 1

Power and references

STEPS 1 TO 6

How the 9 V battery becomes 5 V and 3.3 V, and how the two mid-rails of 2.5 V and 1.65 V are made. Everything else on the board rests on these rails.


Scope out
Input
High-pass
Gain
Low-pass
Driver
ADC
SPI port
Wake-up
Power
Mid-rails

BLOCKS AT WORK IN THIS SECTION (SEE PAGE 3)

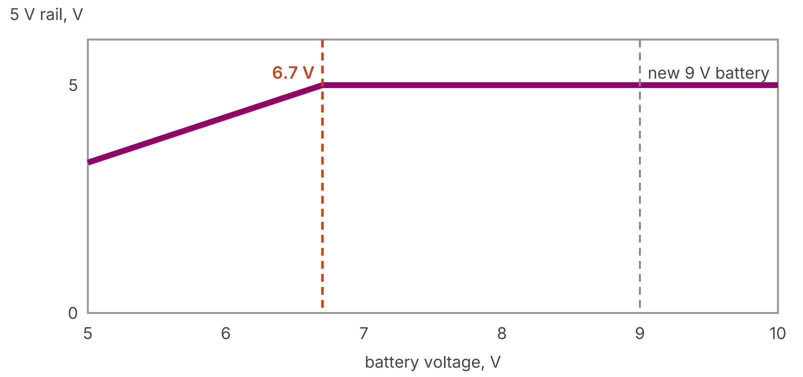
POWER AND REFERENCES

1 How 9 V becomes 5 V

A reversed battery would damage the board. The diode **SS14** (D1) blocks it and costs about 0.4 V. The regulator **AMS1117-5.0** (U7) then needs at least 1.3 V between its input and its output.

So the 5 V rail holds while the battery stays above 6.7 V. Below that, the rail follows the battery down.

5 V RAIL AGAINST THE BATTERY



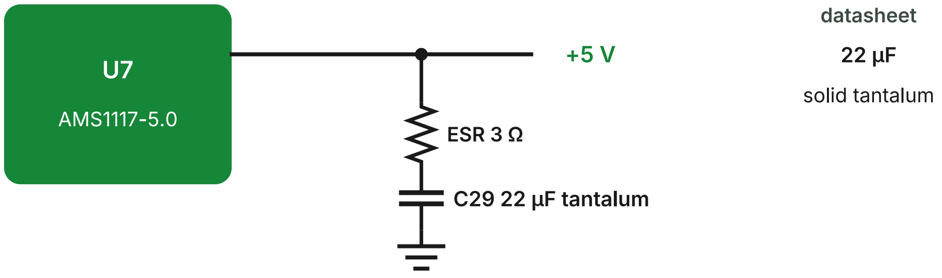
POWER AND REFERENCES

2 Why a tantalum capacitor on 5 V

The AMS1117 datasheet says the output capacitor is part of the regulator loop. It asks for 22 µF of solid tantalum for stability. It gives no ESR range.

A small ceramic has almost no series resistance, and some regulators of this type then oscillate. So C29 is a 22 µF tantalum with about 3 Ω of ESR at 100 kHz.

OUTPUT CAPACITOR



part LCSC C11366, 10 V rating on a 5 V rail

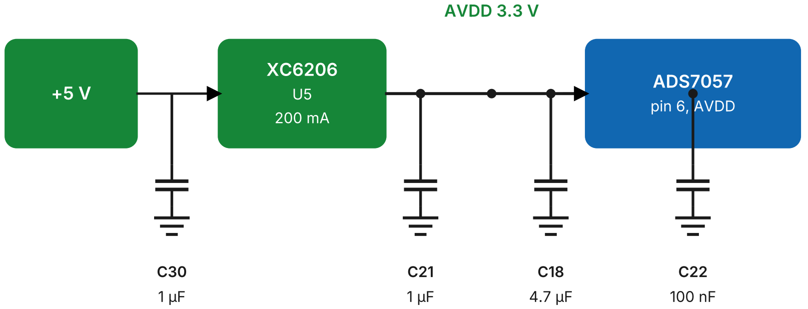
POWER AND REFERENCES

3 Where the ADC reference comes from

The ADS7057 uses its supply pin AVDD as its reference. Full scale is ±AVDD, so noise on AVDD is noise in every code.

A second regulator, the **XC6206** (U5), makes AVDD from the 5 V rail. It accepts ceramic capacitors, so the 1 µF parts C30 and C21 are enough. The 4.7 µF C18 and the 100 nF C22 sit at the ADC pin.

AVDD PATH



ESTIMATE: 0.5 mV rms of noise on AVDD adds about 0.2 LSB to an event of 17 % full scale

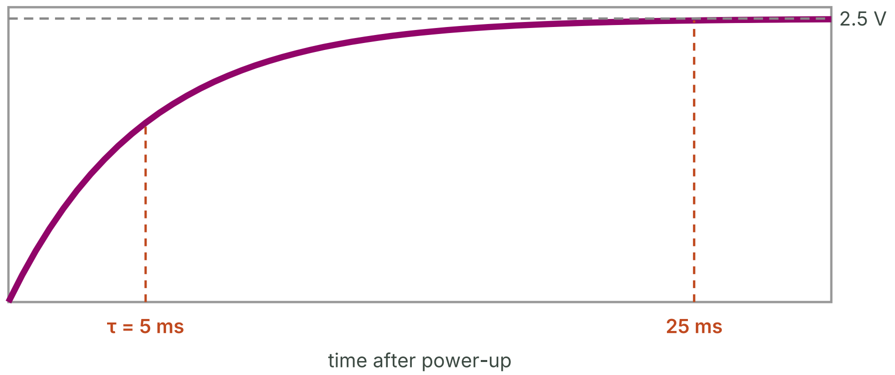
POWER AND REFERENCES

4 How the 2.5 V mid-rail is made

Two 10 kΩ resistors (R7, R8) divide the 5 V rail into 2.5 V. The 1 µF capacitor C7 smooths it. The op-amp U2B then copies this voltage and drives the rail called **VMID5**.

The divider and C7 form an RC of 5 ms, so VMID5 needs about 25 ms to settle after power-up. R9 (10 Ω) and C8 (10 µF) keep the buffer calm.

SETTLING AFTER POWER-UP



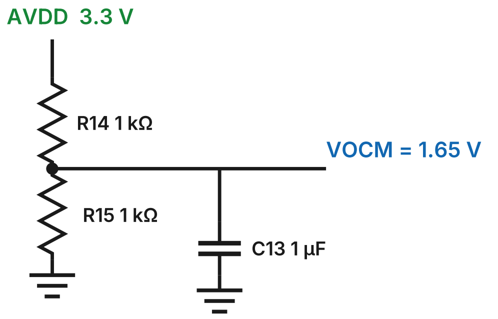
POWER AND REFERENCES

5 How the 1.65 V mid-rail is made

Two 1 kΩ resistors (R14, R15) divide **AVDD** in half, and C13 smooths the result. This voltage, VOCM, tells the THS4531 where to centre its two outputs.

The datasheet asks for the ADC inputs to rest at AVDD/2, within ±0.1 V. The divider follows AVDD by ratio, so any change of AVDD moves VOCM with it.

VOCM



allowed window

1.55 to 1.75 V

AVDD/2 ± 0.1 V

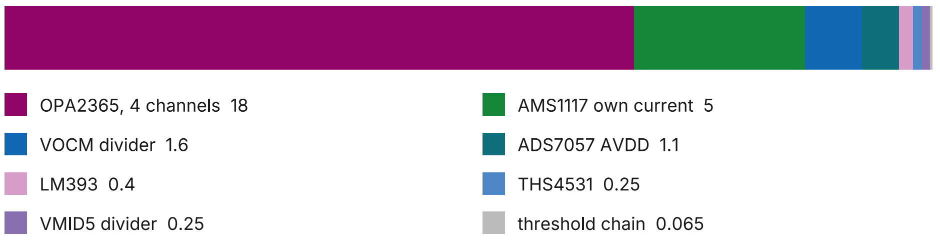
POWER AND REFERENCES

6 How much current, and where to probe

The board draws about 27 mA from the battery (ESTIMATE from datasheet typical values). The four op-amp channels are the largest part, and the AMS1117 uses 5 mA by itself.

At 8.6 V the AMS1117 dissipates about 80 mW. The pads of the capacitors below are the easiest probe points.

ESTIMATE, mA



+5 V	C29 pad 1	**AVDD**	C18 pad 1	**VMID5**	C8 pad 1
VOCM	C13 pad 1	**+3V3**	C20 pad 1	**GND**	J2 pin 2

NOTES

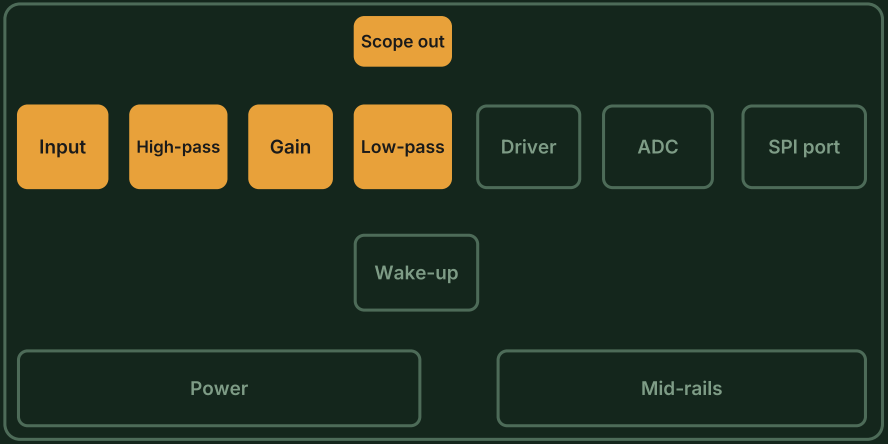
NOTES

SECTION 2

The signal path

STEPS 7 TO 16

From the SMA connector to the output of the low-pass filter: input coupling, high-pass, gain, low-pass, the total response, headroom, the SCOPE OUT tap and the op-amps.


Scope out
Input
High-pass
Gain
Low-pass
Driver
ADC
SPI port
Wake-up
Power
Mid-rails

BLOCKS AT WORK IN THIS SECTION (SEE PAGE 3)

7

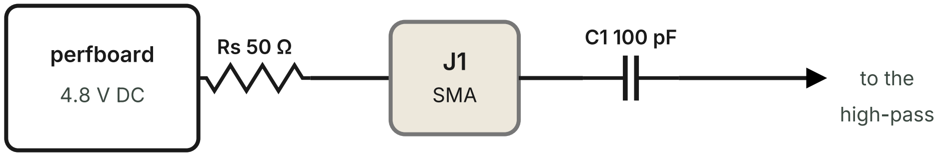
THE SIGNAL PATH

How the signal gets in

The perfboard output reaches J1 through a coax cable. Its own output resistor is 50 Ω. Its output rests at 4.8 V, which is far from the 2.5 V that this board needs.

So the first part is a 100 pF capacitor, C1. It passes the 250 kHz burst and blocks the DC level. A resistor to VMID5 later sets the new DC level.

INPUT


perfboard
4.8 V DC
Rs 50 Ω
J1
SMA
C1 100 pF
to the
high-pass

at 250 kHz, C1 is 6.4 kΩ. At DC it is open.

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THE SIGNAL PATH

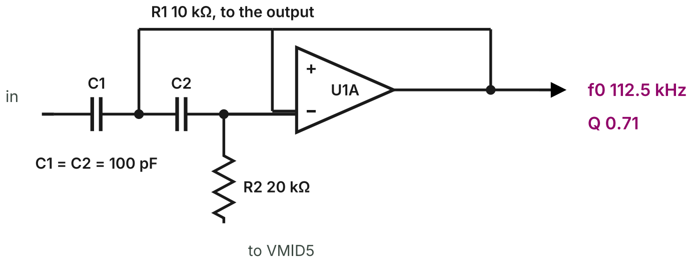
How the high-pass works

The high-pass is a **Sallen-Key** stage: two capacitors in series, two resistors and one op-amp used as a buffer. Low frequencies cannot pass the capacitors.

$$f_0 = \frac{1}{2\pi C\sqrt{R_1 R_2}}, \qquad Q = \frac{1}{2}\sqrt{\frac{R_2}{R_1}}$$

With C = 100 pF, R1 = 10 kΩ and R2 = 20 kΩ, the corner is 112.5 kHz. Q is 0.71. That is a maximally flat response.

SALLEN-KEY HIGH-PASS


R1 10 kΩ, to the output
in
C1
C2
+
U1A
-
C1 = C2 = 100 pF
R2 20 kΩ
to VMID5
f0 112.5 kHz
Q 0.71

9

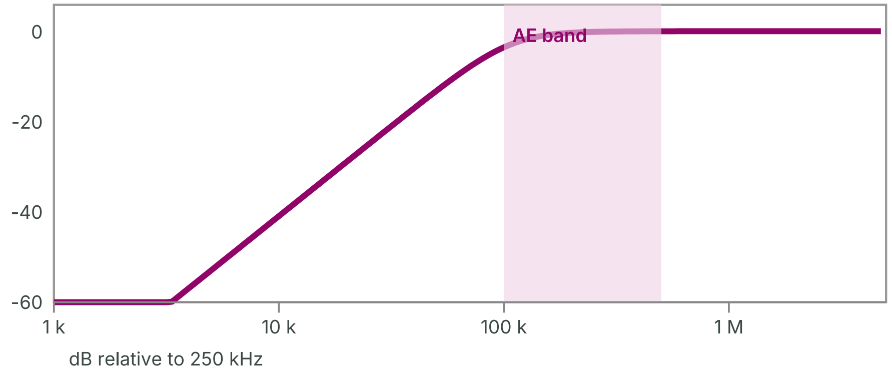
THE SIGNAL PATH

Why the corner is at 113 kHz

Most noise on the perfboard sits below 100 kHz, from 11.5 to 26.8 mV rms. The AE band starts at 100 kHz and holds only about 4 mV rms.

So the high-pass sits just under the band. At 10 kHz it already removes 41 dB of this stage alone. A corner any higher would also cut the events.

SIMULATION, HIGH-PASS STAGE ONLY



10

THE SIGNAL PATH

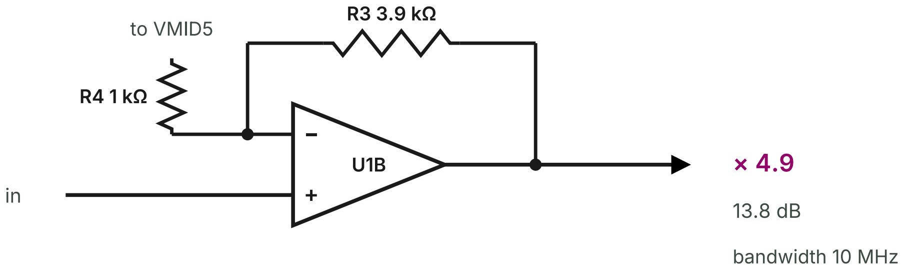
How the gain is set

The second half of U1 is a non-inverting amplifier. Its gain comes from two resistors: R3 from the output back to the minus input, and R4 from that input to VMID5.

$$G = 1 + \frac{R_3}{R_4} = 1 + \frac{3.9\,\mathrm{k\Omega}}{1\,\mathrm{k\Omega}} = 4.9$$

Because R4 goes to VMID5 and not to ground, the gain is 1 for DC. The stage output rests at 2.5 V, and the gain applies only to the burst.

GAIN 4.9


to VMID5
R3 3.9 kΩ
R4 1 kΩ
-
U1B
+
in
× 4.9
13.8 dB
bandwidth 10 MHz

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THE SIGNAL PATH

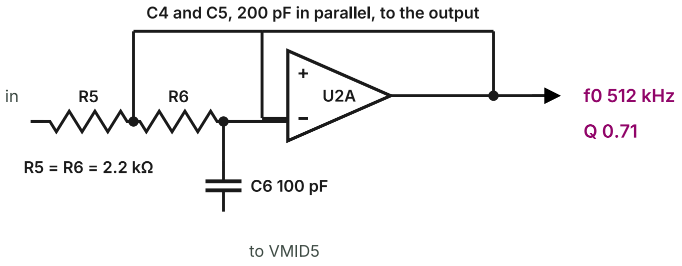
How the low-pass works

The low-pass is also a Sallen-Key stage, with the parts swapped: two resistors in series and two capacitors. The capacitors give high frequencies a path to VMID5.

$$f_0 = \frac{1}{2\pi R\sqrt{C_1C_2}}, \qquad Q = \frac{1}{2}\sqrt{\frac{C_1}{C_2}}$$

With R = 2.2 kΩ, C1 = 200 pF (C4 and C5 in parallel) and C2 = 100 pF (C6), the corner is 512 kHz. Q is 0.71.

SALLEN-KEY LOW-PASS


C4 and C5, 200 pF in parallel, to the output
in
R5
R6
U2A
+
−
f0 512 kHz
Q 0.71
R5 = R6 = 2.2 kΩ
C6 100 pF
to VMID5

12

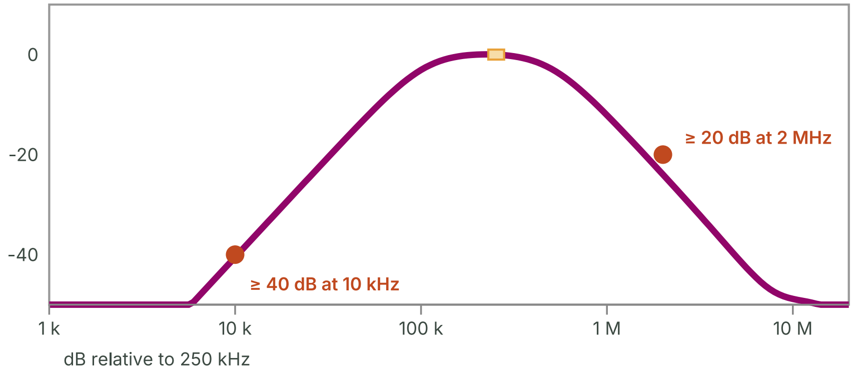
THE SIGNAL PATH

The total response

Put the three stages together and add the 50 Ω source. The simulation, with TI's own OPA365 model, gives 12.3 dB of gain at 250 kHz. That is ×4.1, a little below the ideal ×4.9.

The loss at 10 kHz is 40.5 dB against a target of 40 dB. With part tolerances 11 of 150 runs fall under 40 dB, the worst at 39.7 dB.

SIMULATION, SMA TO ADC



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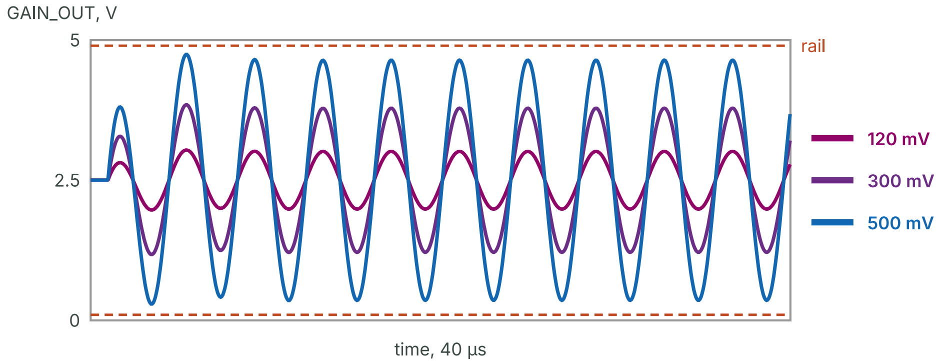
THE SIGNAL PATH

What clips first

The 5 V stages are the narrowest part of the path. They can swing about 2.4 V either side of 2.5 V. A burst clips there when it is larger than 0.54 V at the perfboard output.

A typical recorded event is 120 mV, so the margin is 4.5×. At the ADC that event fills 16 % of the full scale.

SIMULATION, 3 BURSTS



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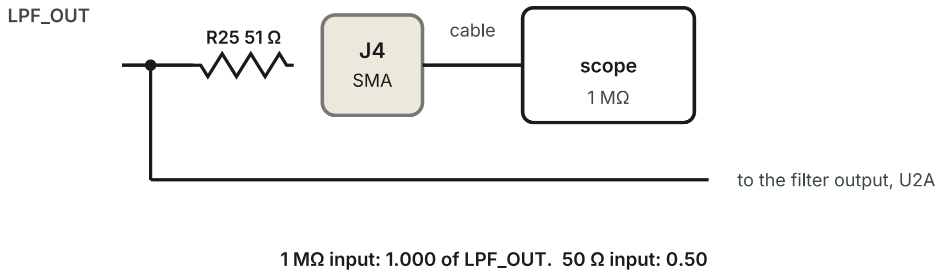
THE SIGNAL PATH

What SCOPE OUT is for

J4 gives the scope the same signal that the ADC sees, taken at the output of the low-pass. A 51 Ω resistor (R25) sits between the op-amp and the connector.

The resistor keeps the cable capacitance away from the op-amp, so the filter cannot ring. Use the 1 MΩ input of the scope. A 50 Ω input would form a divider with R25 and halve the signal.

CABLE LOAD


LPF_OUT
R25 51 Ω
J4
SMA
cable
scope
1 MΩ
to the filter output, U2A
1 MΩ input: 1.000 of LPF_OUT. 50 Ω input: 0.50

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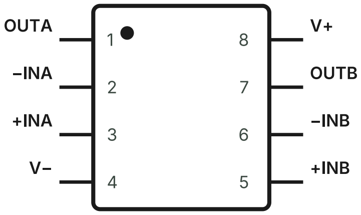
THE SIGNAL PATH

What the OPA2365 does here

Two dual op-amps, U1 and U2, do all the active filtering. The OPA2365 is rail-to-rail on input and output and runs from 2.2 to 5.5 V. Its gain-bandwidth is 50 MHz and its noise is 4.5 nV/√Hz.

Each package holds two channels. The table shows which stage uses which channel.

SOIC-8, TOP VIEW



U1A pins 1 2 3	high-pass buffer
U1B pins 5 6 7	gain × 4.9
U2A pins 1 2 3	low-pass buffer
U2B pins 5 6 7	VMID5 buffer

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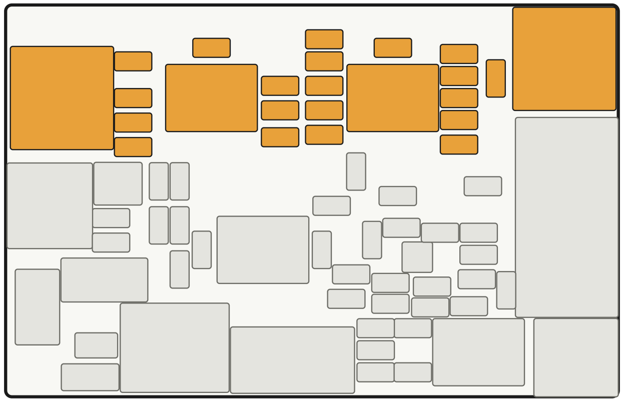
THE SIGNAL PATH

Where these parts are

The signal parts sit in the top half of the board, from J1 on the left to U2 on the right. U1 holds the high-pass and the gain. U2 holds the low-pass and the VMID5 buffer.

J4 and its resistor R25 stand in the top right corner, away from the ADC.

TOP VIEW, ACTIVE PARTS IN AMBER



Input and high-pass
C1 C2 R1 R2

Gain
R3 R4

Low-pass
R5 R6 C4 C5 C6

Op-amps
U1 U2 C3 C10

VMID5
R7 R8 C7 R9 C8 C9

Connectors
J1 J4 R25

NOTES

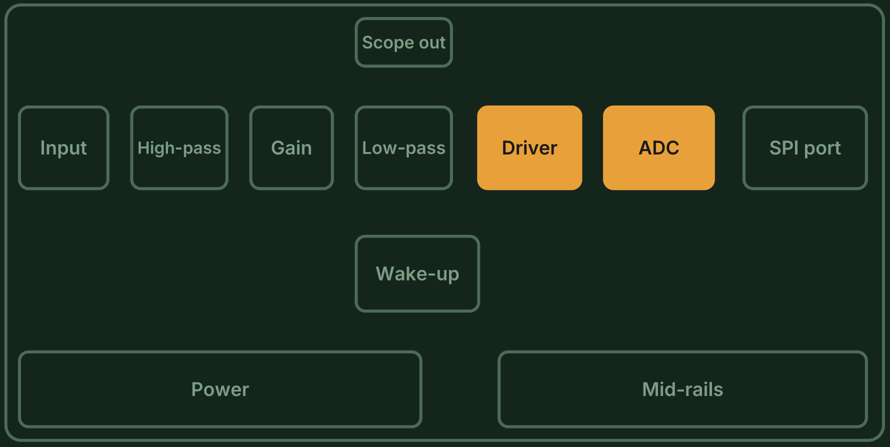
NOTES

SECTION 3

Driver and ADC

STEPS 17 TO 24

How one wire becomes two, how the ADC input is protected from its own sampling kick, what the ADS7057 is and what its numbers mean for this board.


Scope out
Input
High-pass
Gain
Low-pass
Driver
ADC
SPI port
Wake-up
Power
Mid-rails

BLOCKS AT WORK IN THIS SECTION (SEE PAGE 3)

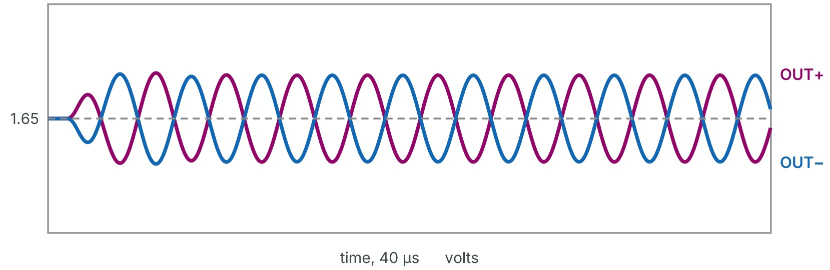
17
DRIVER AND ADC
How one wire becomes two

The ADS7057 needs two inputs that move in opposite directions. The **THS4531** (U3) makes them from one wire. The capacitor C11 and R10 bring the signal in. The corner of this coupling is 159 Hz, far below the band.

$$V_{\mathrm{OUT+}} - V_{\mathrm{OUT-}} = \frac{R_{12}}{R_{10}} V_{\mathrm{IN}} = 1 \cdot V_{\mathrm{IN}}$$

Each output moves half of the signal around VOCM, in opposite directions.

SIMULATION, 120 mV BURST



18
DRIVER AND ADC
What to know about the THS4531

Three facts from the datasheet shape the board. The **PD** pin must be driven, so U3 ties it to 5 V. The **VOCM** pin accepts 0.95 to 4.0 V, so 1.65 V is fine.

The third fact is a limit. The designers of the part expect about 2 pF of load between its outputs. A larger load needs a series resistor.

DATASHEET AGAINST THIS BOARD

item	datasheet	this board	
Supply	2.5 to 5.5 V	**5 V**	
Gain-bandwidth	27 MHz	**gain 1**	
Noise	10 nV/√Hz	**0.2 LSB**	
VOCM range at 5 V	0.95 to 4.0 V	**1.65 V**	
PD pin	must be driven	**tied to 5 V**	
Load between outputs	2 pF nominal	**400 pF + 20 Ω**	OPEN

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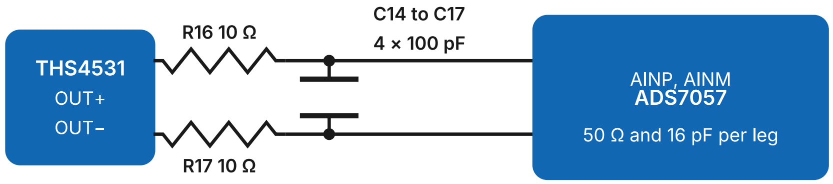
DRIVER AND ADC
What the 10 Ω and 400 pF are for

When the ADC samples, its 16 pF capacitor connects to the input and pulls a burst of charge. A step of 0.56 V moves 9 pC. A big capacitor next to the pins supplies this charge, so the driver sees only a small glitch.

Four 100 pF capacitors make 400 pF, which is 25 times the sampling capacitor. The datasheet asks for at least 20 times. The two 10 Ω resistors isolate the driver. The pole is at 20 MHz, far above the band.

KICKBACK FILTER



the node dips by 22 mV (53 LSB) and recovers in 54 ns at worst

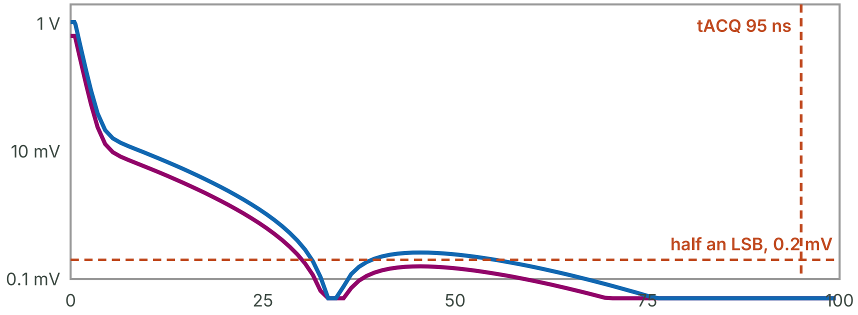
the 25× rule: 16 pF × 20 = 320 pF minimum

20
DRIVER AND ADC
What the ADC input looks like

The sampling switch of the ADC is 50 Ω in series with 16 pF on each leg. The simulation closes the switch on a signal that was left at the previous sample. The input settles to half an LSB in 29 ns for a typical step and 54 ns for a large one.

The ADC needs the input settled within 95 ns at the highest rate. So the margin is a factor of about two on the worst case.

SIMULATION, SETTLING



ns after the switch closes. Error against the final value, log scale.

21

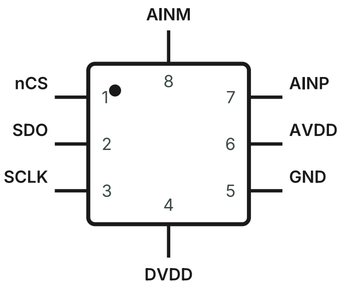
DRIVER AND ADC

What the ADS7057 is

A 14-bit successive-approximation ADC in a 1.5 × 1.5 mm package with eight pads. It samples on the falling edge of CS and sends its result over SPI.

The table lists the numbers that matter for this board. All come from datasheet SBAS821.

X2QFN-8, TOP VIEW



Resolution	14 bit, no missing codes
Rate	up to 2.5 MSPS
Supply	AVDD 2.35 to 3.6 V, DVDD 1.65 to 3.6 V
Input	differential, ±AVDD, common mode AVDD/2
SNR	79.6 dB at 2 kHz
INL, DNL	±0.9 LSB, ±0.3 LSB typical
SPI	mode 1, SCLK up to 60 MHz
Conversion	18 SCLK, then at least 95 ns acquisition

22

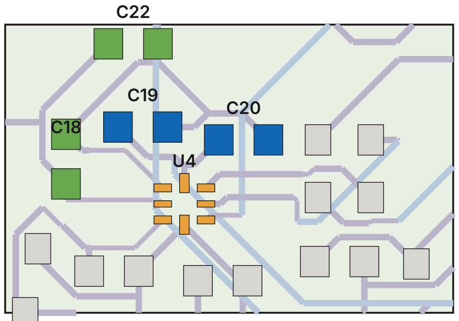
DRIVER AND ADC

How the ADC is decoupled

AVDD gets a 4.7 µF (C18) and a 100 nF (C22). DVDD gets a 1 µF (C19) and a 100 nF (C20). The datasheet asks for 3.3 µF and 1 µF and says to keep them close, with no vias between the pin and the capacitor.

On this board C22 is 4.4 mm from the ADC and C18 is 3.0 mm. C20 is 1.6 mm and C19 is 1.8 mm. The 100 nF parts sit farther than the large ones, which is a point to improve.

REAL BOARD, 7 ×



AVDD parts green, DVDD parts blue

23

DRIVER AND ADC

How a voltage becomes a code

The 120 mV event of the simulation reaches 0.517 V between AINP and AINM. The code is the voltage divided by 1 LSB, rounded down.

$$\text{code} = \left\lfloor \frac{V_{\text{AINP}} - V_{\text{AINM}}}{1\ \text{LSB}} \right\rfloor$$

So the peak is 1284, which is 0x0504. The negative peak -0.507 V gives -1258, which is 0x3B16 in two's complement.

FROM THE SIMULATION

D13 is the sign bit

positive peak	0	0	0	1	0	1	0	0	0	0	0	1	0	0	0x0504
negative peak	1	1	1	0	1	1	0	0	0	1	0	1	1	0	0x3B16

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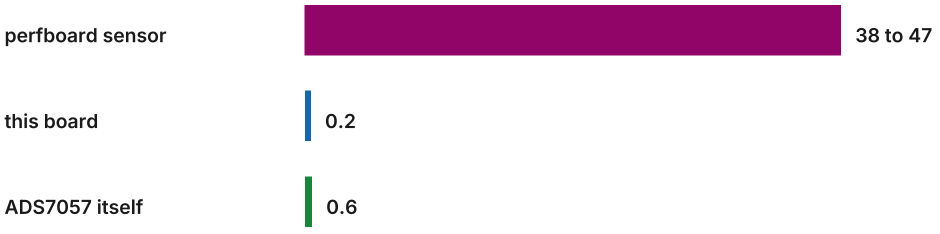
DRIVER AND ADC

How much noise the ADC sees

Three noise sources add. The sensor noise of the perfboard, through this board, is 38 to 47 LSB rms. The board adds 0.2 LSB. The ADC itself adds about 0.6 LSB.

So the sensor decides the noise floor. The board and the ADC are small by comparison, and this is what a test board should show.

LSB RMS, ESTIMATE FOR THE ADC

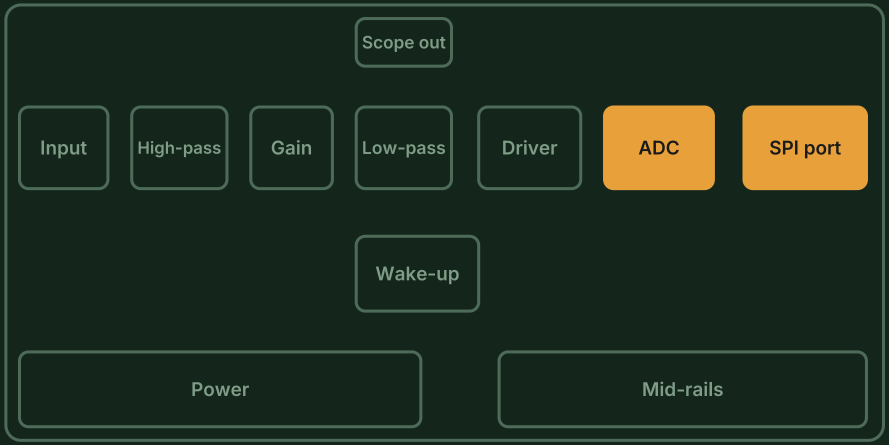


SECTION 4

The SPI link

STEPS 25 TO 30

The four wires from the ADC to the ZedBoard, what one frame looks like, why the data comes one frame late, the calibration at power-up and how fast the link can go.


Scope out
Input
High-pass
Gain
Low-pass
Driver
ADC
SPI port
Wake-up
Power
Mid-rails

BLOCKS AT WORK IN THIS SECTION (SEE PAGE 3)

25

THE SPI LINK

Which pin is which on the PMOD

The board plugs into one PMOD port of the ZedBoard, JA to JD. The board uses eight of the twelve pins. **CS**, **SCLK** and **SDO** carry SPI. **nWAKE** is the wake-up line. 3V3 powers the ADC logic, and GND is the return.

There is no MOSI wire, because this ADC only sends data. The numbering follows the Digilent SPI PMOD, so pin 1 is CS.

J3, SEEN FROM THE TOP

left column	right column
1 CS	7 not used
2 not used	8 not used
3 SDO	9 not used
4 SCLK	**10 nWAKE**
5 GND	**11 GND**
6 3V3	**12 3V3**

The number is the PMOD pin. Row 1 holds pin 1 and pin 7.

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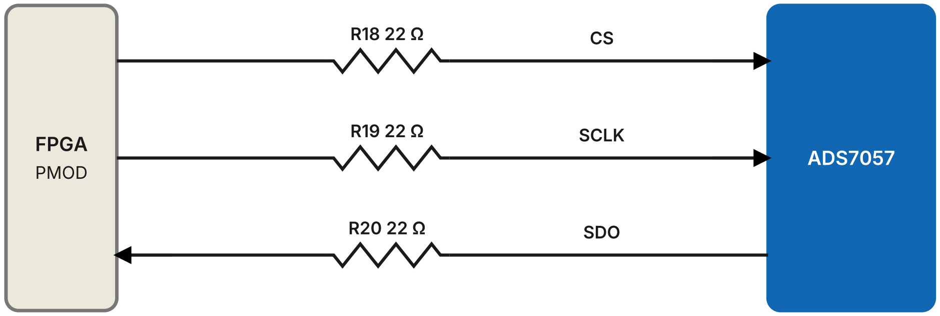
THE SPI LINK

Why 22 Ω in each line

R18, R19 and R20 sit in the CS, SCLK and SDO lines at the ADC. Together with the small capacitance of the trace and the pin they slow the edge of the FPGA output a little and damp ringing.

The cost in speed is tiny. 22 Ω with about 10 pF (ESTIMATE) gives 0.2 ns, and one bit at 60 MHz lasts 16.7 ns.

SERIES RESISTOR


FPGA
PMOD
R18 22 Ω
CS
R19 22 Ω
SCLK
R20 22 Ω
SDO
ADS7057

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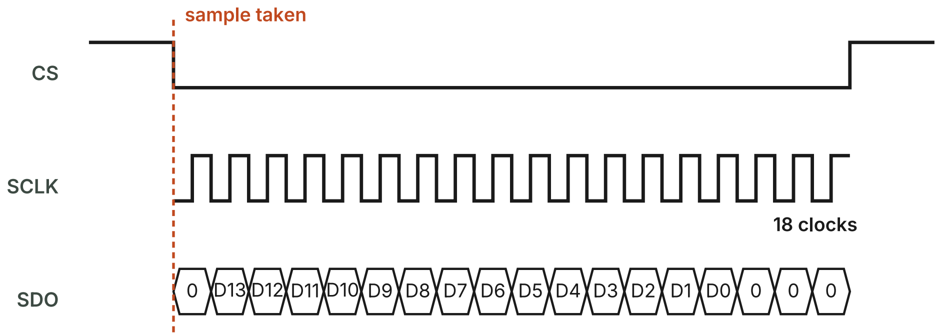
THE SPI LINK

What one frame looks like

A **frame** starts when CS falls. The ADC samples at that moment. The host then sends 18 clock pulses. On SDO the ADC answers with a leading 0, then 14 data bits, most significant bit first, then three zeros.

After the 18th clock the ADC converts nothing more. CS goes high and the ADC acquires the next sample for at least 95 ns.

ONE FRAME


sample taken
CS
SCLK
18 clocks
SDO
0
D13
D12
D11
D10
D9
D8
D7
D6
D5
D4
D3
D2
D1
D0
0
0
0

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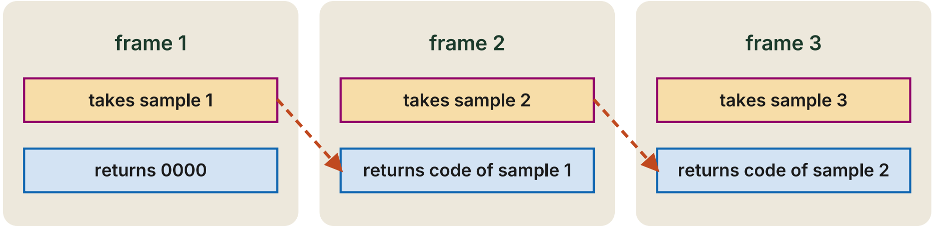
THE SPI LINK

Why the data is one frame late

The ADC uses the clocks of a frame to send the result of the **previous** conversion. The conversion of the current sample runs inside the same frame, and its result waits for the next one.

So the first frame after power-up returns all zeros, and every later frame returns the sample that was taken one frame before.

THREE FRAMES


frame 1
takes sample 1
returns 0000
frame 2
takes sample 2
returns code of sample 1
frame 3
takes sample 3
returns code of sample 2

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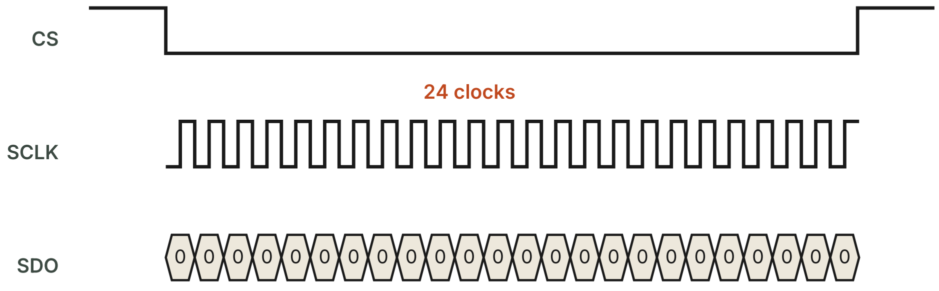
THE SPI LINK

What happens at power-up

The datasheet recommends an offset calibration at power-up. The host does it by sending **24** clocks in the first frame, not 18. The ADC then disconnects its input, corrects its own offset and sends zeros.

At a 20 MHz clock this first frame takes 1200 ns plus 95 ns of acquisition, so 1295 ns. Repeat it when AVDD or the temperature changes a lot.

FIRST FRAME



zeros while the offset is calibrated

30

THE SPI LINK

How fast the link can go

One sample costs 18 clock periods plus 95 ns. So the sample rate is set by SCLK. The datasheet allows up to 60 MHz, which gives the full 2.5 MSPS.

There are no level shifters on this board, so the FPGA drives the ADC pins directly at 3.3 V. Start at 10 to 20 MHz and raise the clock when the first test works.

SAMPLE RATE AGAINST SCLK

SCLK	one sample	sample rate
10 MHz	1895 ns	**0.53 MSPS**
20 MHz	995 ns	**1.01 MSPS**
25 MHz	815 ns	**1.23 MSPS**
40 MHz	545 ns	**1.83 MSPS**
60 MHz	395 ns	**2.50 MSPS**

NOTES

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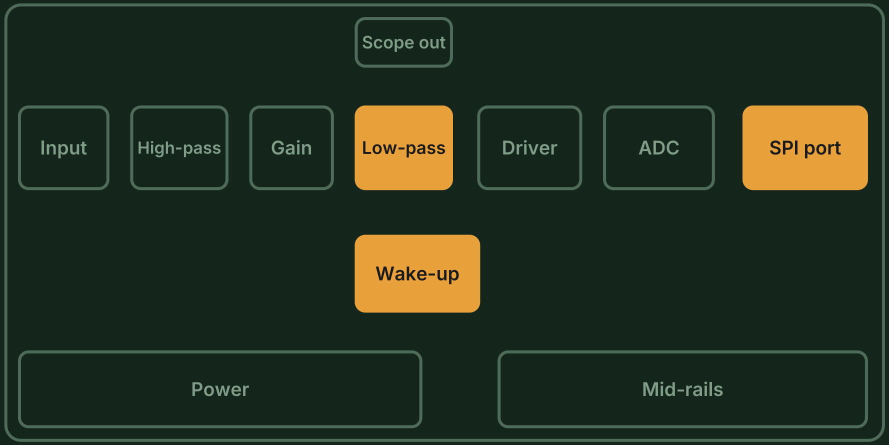
SECTION 5

The wake-up

STEPS 31 TO 35

How the LM393 turns a strong burst into a low level on nWAKE, how the trimmer sets the window and how to set it on the bench.


Scope out
Input
High-pass
Gain
Low-pass
Driver
ADC
SPI port
Wake-up
Power
Mid-rails

BLOCKS AT WORK IN THIS SECTION (SEE PAGE 3)

31

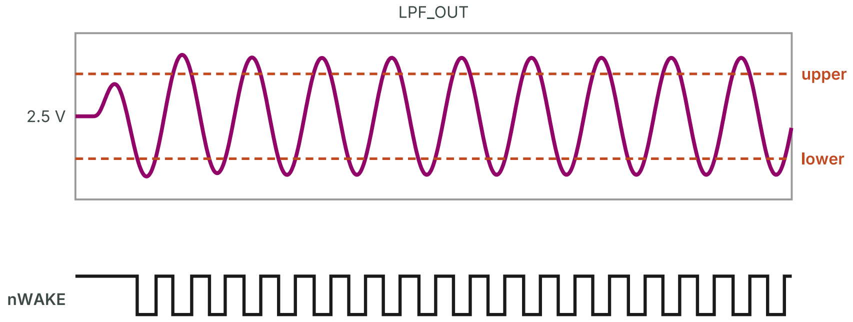
THE WAKE-UP

What a window comparator is

The LM393 holds two comparators. One trips when the signal rises above the **upper** threshold. The other trips when it falls below the **lower** threshold. Both thresholds sit around 2.5 V.

The two outputs share one wire. So nWAKE goes low as soon as the signal leaves the window on either side.

SIMULATION, 120 mV BURST



window ±357 mV, trimmer at 10 kΩ

32

THE WAKE-UP

How the window is set

One resistor chain from 5 V to ground makes both thresholds. The trimmer RV1 and a 1 kΩ resistor set the distance between them.

$$V_{\text{half}} = 2.5\,\text{V} \cdot \frac{1\,\text{k}\Omega + R_p}{67\,\text{k}\Omega + R_p}$$

The last column converts the window back to the perfboard output with a gain of 4.1.

RP AGAINST WINDOW

Rp	upper	lower	half window	at J1
0 kΩ	2.537 V	2.463 V	**37 mV**	**9 mV**
5 kΩ	2.708 V	2.292 V	**208 mV**	**51 mV**
10 kΩ	2.857 V	2.143 V	**357 mV**	**87 mV**
14 kΩ	2.963 V	2.037 V	**463 mV**	**113 mV**
20 kΩ	3.103 V	1.897 V	**603 mV**	**147 mV**

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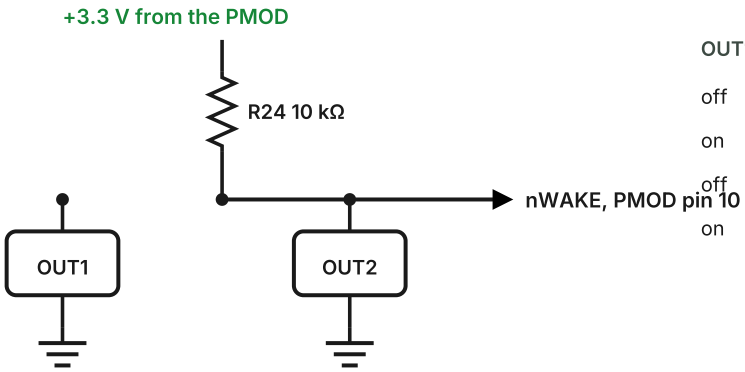
THE WAKE-UP

Why nWAKE is open-collector

The LM393 output can only pull the line to ground. It cannot drive it high. A 10 kΩ resistor (R24) pulls the line up to 3.3 V from the PMOD.

So the FPGA never sees 5 V, although the LM393 runs from 5 V. The two outputs share one wire, so either one can pull nWAKE low.

WIRED-OR



OUT1	OUT2	nWAKE
off	off	**high**
on	off	**low**
off	on	**low**
on	on	**low**

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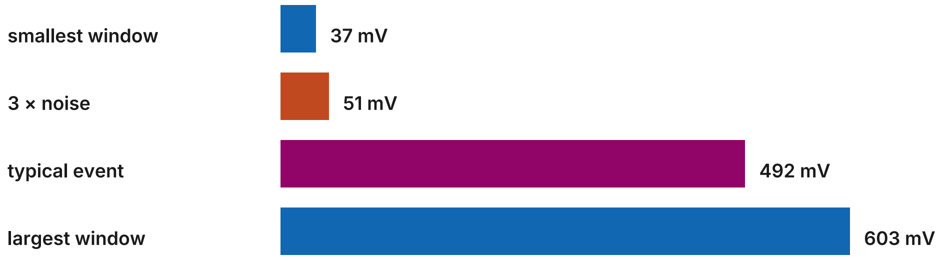
THE WAKE-UP

How fast and how accurate

The LM393 answers in about 1.3 µs. That is slow for a clock but fast for a wake-up. Its input offset is about 5 mV, which is small against the window.

The noise decides the smallest useful setting. The quiet baseline of 4.1 mV rms becomes 17 mV rms at the comparator. Three times that is 51 mV. Below Rp = 0.4 kΩ the noise alone trips the output.

NOISE AGAINST WINDOW, ESTIMATE



35

THE WAKE-UP

How to set the trimmer

Choose the burst that should wake the system. Feed that burst to J1. Watch SCOPE OUT and nWAKE together. Turn RV1 slowly until nWAKE drops for that burst and stays high without it.

A 50 mV burst at J1 needs Rp near 4.9 kΩ. A 100 mV burst needs Rp near 12.0 kΩ. The trimmer is a 20 kΩ part, so the middle of its travel is close to 100 mV.

BENCH STEPS

1. **Feed 250 kHz, 100 mV to J1**
2. **Read 0.41 V peak on SCOPE OUT**
3. **Turn RV1 until nWAKE just pulses**
4. **Remove the burst. nWAKE must stay high**

clockwise should widen the window. Check with a meter on VTH_H

NOTES

NOTES

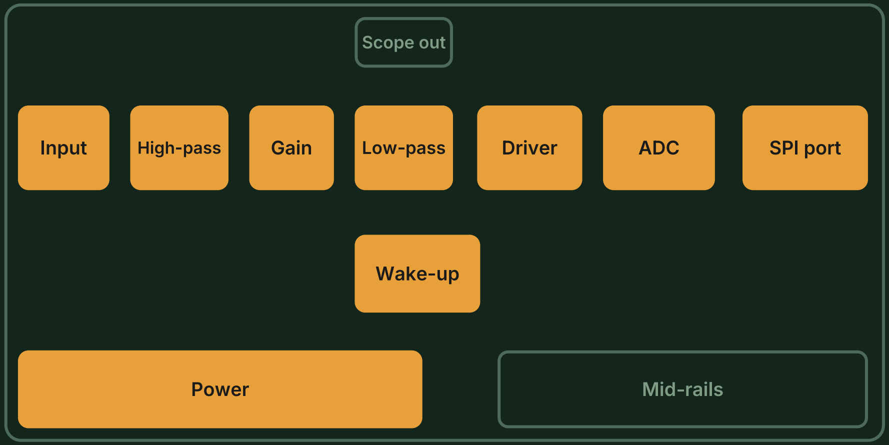
NOTES

SECTION 6

The board

STEPS 36 TO 44

The stack, the floor plan, where every part sits, how the tracks and the power are routed, what the silkscreen says and which footprints need care.



Scope out

Input

High-pass

Gain

Low-pass

Driver

ADC

SPI port

Wake-up

Power

Mid-rails

BLOCKS AT WORK IN THIS SECTION (SEE PAGE 3)

36

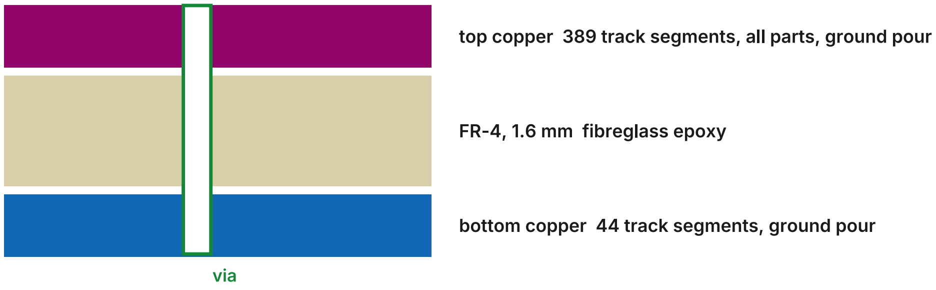
THE BOARD

The stack and the size

The board is 50 × 32 mm with two copper layers on 1.6 mm FR-4. Both layers carry a ground pour. 82 of the 107 vias are ground stitching vias that tie the two pours together.

All 433 track segments are 0.2 mm wide, except 55 short segments of 0.15 mm near the fine-pitch pads. Vias are 0.6 mm with a 0.3 mm drill.

FROM THE BOARD FILE


top copper 389 track segments, all parts, ground pour
FR-4, 1.6 mm fibreglass epoxy
bottom copper 44 track segments, ground pour
via

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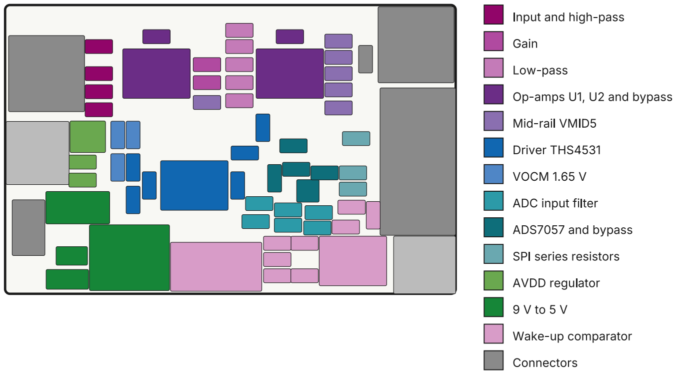
THE BOARD

How the floor is planned

The signal runs left to right across the top half, as in the block diagram. The driver and the ADC sit in the middle. The power parts take the bottom left. The comparator sits in the bottom right.

The PMOD connector is on the right edge so the board hangs on the ZedBoard. The SMA connectors stand at the top, away from the ADC.

TOP VIEW, COLORS BY BLOCK


Input and high-pass
Gain
Low-pass
Op-amps U1, U2 and bypass
Mid-rail VMID5
Driver THS4531
VOCM 1.65 V
ADC input filter
ADS7057 and bypass
SPI series resistors
AVDD regulator
9 V to 5 V
Wake-up comparator
Connectors

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THE BOARD

Where every part is

The table lists the blocks, the references and the count. The board has 69 footprints: 61 surface-mount parts, 5 through-hole parts and 2 mounting holes, plus one not-fitted capacitor.

Every part has a place in exactly one block, and the same reference appears in the schematic, the BOM and the placement file.

BLOCKS, REFERENCES

🟪	**Input and high-pass**	C1 C2 R1 R2
🟪	**Gain**	R3 R4
🟪	**Low-pass**	C4 C5 C6 R5 R6
🟪	**Op-amps U1, U2 and bypass**	C3 C10 U1 U2
🟪	**Mid-rail VMID5**	C7 C8 C9 R7 R8 R9
🟦	**Driver THS4531**	C11 C12 R10 R11 R12 R13 U3
🟦	**VOCM 1.65 V**	C13 R14 R15
🟦	**ADC input filter**	C14 C15 C16 C17 R16 R17
🟦	**ADS7057 and bypass**	C18 C19 C20 C22 U4
🟦	**SPI series resistors**	R18 R19 R20
🟩	**AVDD regulator**	C21 C30 U5
🟩	**9 V to 5 V**	C23 C29 D1 U7
🟪	**Wake-up comparator**	C26 C27 C28 C31 RV1 R21 R22 R23 R24 U6
⬛	**Connectors**	J1 J2 J3 J4 R25
⬜	**Holes**	H1 H2

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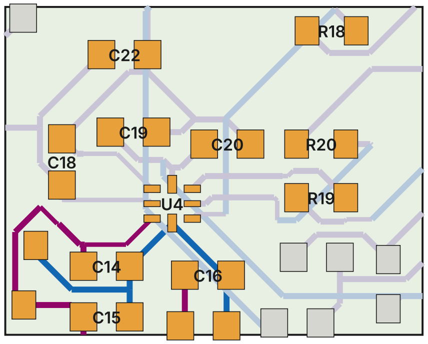
THE BOARD

Why the ADC corner is built this way

Everything that touches the ADC input sits around U4: the two 10 Ω resistors, the four 100 pF capacitors, the supply capacitors and the three 22 Ω resistors. The tracks AINP and AINM are 17 and 12 mm long.

The 5 mm difference is harmless here. A 5 mm track delays a signal by about 30 ps, and a 500 kHz wave lasts 2 µs.

REAL BOARD, ADC CORNER


R18
C22
C19
C20
R20
C18
U4
R19
C14
C16
C15

40

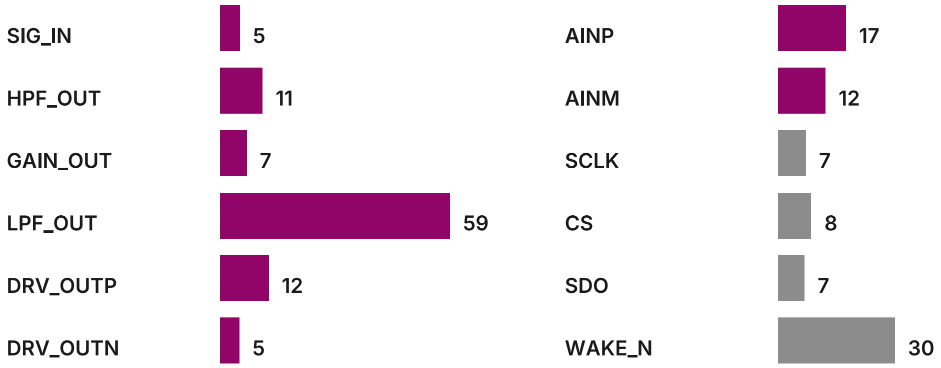
THE BOARD

How the analog tracks are routed

The longest analog net is LPF_OUT at 59 mm, because it feeds the SCOPE OUT connector, the comparator and the driver. All other analog nets are under 20 mm.

The router chose the paths, and the board is small enough that none of them matters at 500 kHz. The figure lists the lengths.

NET LENGTH, mm



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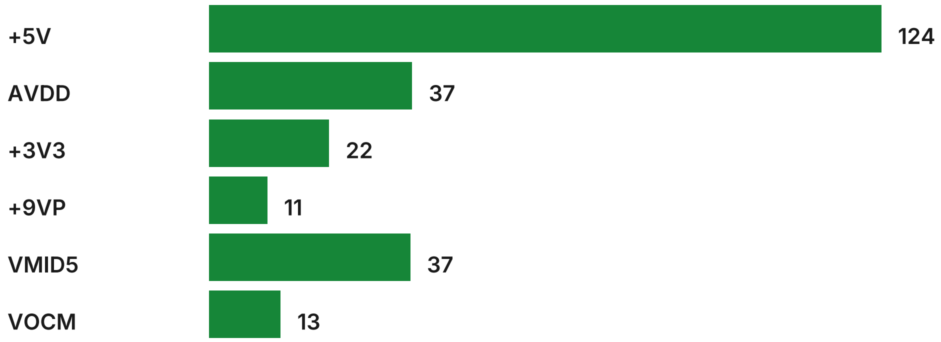
THE BOARD

How power and ground are routed

Ground is a pour on both layers, tied by 82 vias, so every ground pad finds a short path. The power nets are ordinary 0.2 mm tracks, not wide ones, because the currents are small.

The +5 V net is 124 mm of track in total. With 22 mA this drops about 7 mV across its whole length (ESTIMATE, worst case).

POWER NET LENGTH, mm



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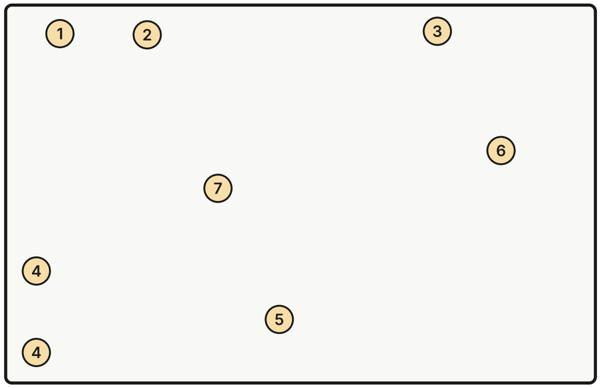
THE BOARD

What each label on the silkscreen means

The front carries the labels that a user needs. The names of the parts (R16, C29 and so on) are not on the silkscreen, because the board is dense. They are in the KiCad file and in the 1:1 print.

The back carries the PMOD pin list and the title.

FRONT SIDE



1 SENSOR IN: J1, from the perfboard
2 CH box: write the channel number
3 SCOPE OUT: J4, the signal at the ADC
4 +9V and GND: battery, square pad is +
5 WAKE LEVEL: trimmer RV1
6 1: PMOD pin 1
7 logo: Auselia mark

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THE BOARD

Which footprints are tricky

The ADS7057 pads are 0.55 × 0.25 mm at 0.5 mm pitch. The gap between pads is 0.25 mm, so the solder mask web is thin. JLCPCB assembled it in the quote without a warning.

Four other parts needed care. The factory model of each one had a different orientation or origin from the footprint, and the CPL now corrects it.

PARTS TO WATCH

U4 ADS7057
X2QFN-8, 1.5 mm
pin 1 mark checked

U5 XC6206
SOT-23, 3 pins
rotation 180

C29 tantalum
polarity: + on pad 1
rotation 0

J3 PMOD
right-angle 2 × 6
5.9 mm beyond the edge

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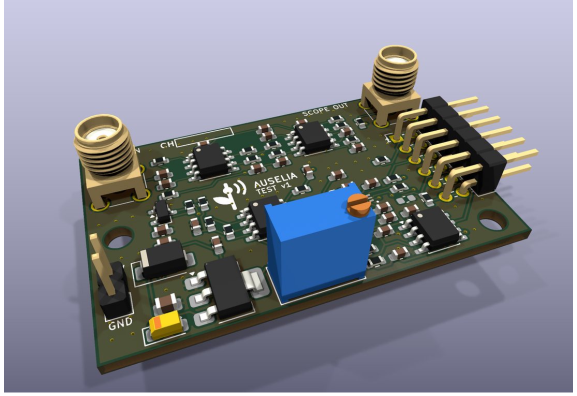
THE BOARD

What the board looks like

The render is from the final KiCad file. J1 and J4 stand on the top left and right. The PMOD header J3 points off the right edge. The blue trimmer RV1 is at the bottom.

The two holes are M3 mounting holes.

3D RENDER, KICAD



J1: SMA in, top left

J4: SCOPE OUT, top right

J3: PMOD, right edge

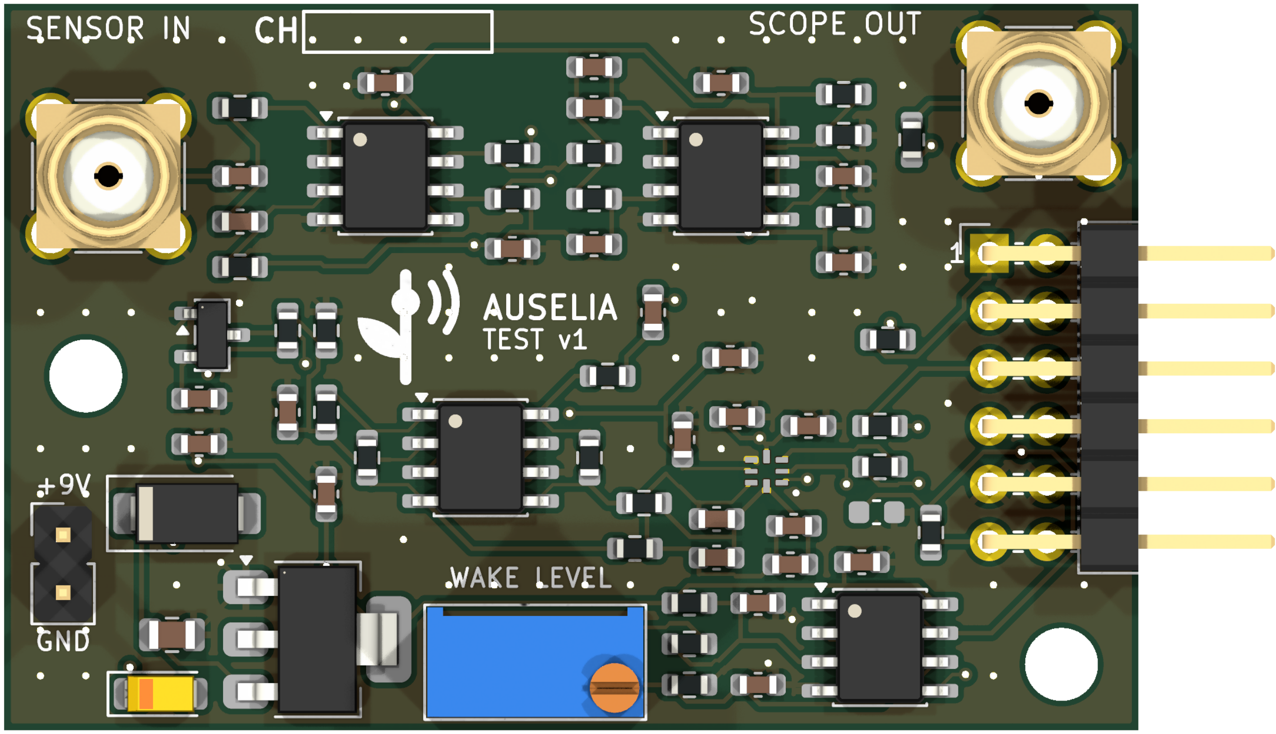
RV1: wake level trimmer

U4: ADS7057, centre right

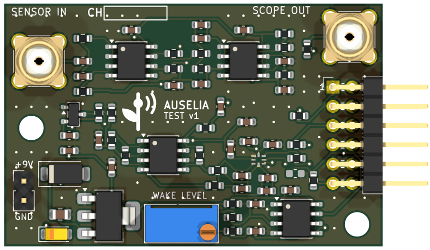
NOTES

NOTES

NOTES


SENSOR IN
CH
SCOPE OUT
1
AUSELIA
TEST v1
+9V
GND
WAKE LEVEL

10 mm at this scale (3 : 1) Board 50.0 × 32.0 mm, KiCad 3D render, view from the top


SENSOR IN
CH
SCOPE OUT
AUSELIA
TEST v1
+9V
GND
WAKE LEVEL

The same view at real size, 1 : 1

SECTION 7

Making the board

STEPS 45 TO 48

The files the factory needs, the parts list, what the placement viewer corrected and what the order costs.

45

MAKING THE BOARD

Which files go to the factory

JLCPCB needs three files. The Gerber zip makes the bare board. The BOM names every part. The CPL gives the position and angle of each part.

The zip holds 13 files. The BOM has 27 lines and 66 parts. The CPL has 66 rows. The two mounting holes and the not-fitted capacitor C31 are in none of the two lists.

FILES IN hw/fab

ADC_Test_gerbers.zip
Copper, mask, silk, paste, outline, drill

ADC_Test_BOM_JLC.csv
Value, references, footprint, LCSC number

ADC_Test_CPL_JLC.csv
Reference, X, Y, layer, rotation

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MAKING THE BOARD

What is in the BOM

27 lines hold 66 parts. Four lines hold extended parts, and JLCPCB charges a fee for each type. All other lines hold basic parts.

The ADC, the driver and the two op-amps are the extended parts. The tantalum capacitor is the fourth.

EXTENDED PARTS, PRICE PER PART

Ref	Part	LCSC	Price
U1,U2	OPA2365	C130011	$1.38
U3	THS4531	C705347	$2.99
U4	ADS7057	C2669902	$7.98
C29	22uF 10V tantalum	C11366	$0.19

Through-hole parts (5 lines): J1 J4 SMA, J2, J3, RV1. JLCPCB solders them by hand, for a fee per joint.

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MAKING THE BOARD

What the placement viewer corrected

The JLCPCB viewer showed four parts turned wrong. The KiCad footprints and the JLC library disagree on the zero angle. Each fix sits in `jlc_files.py`, not in the board.

The trimmer RV1 also moved 2.54 mm in X and 1 mm in Y, because its library origin differs from the pin 1 pad.

KICAD ROTATION AGAINST CPL ROTATION

Ref	KiCad	CPL	Why
J3	0°	90°	viewer showed it turned; set absolute angle

Rule: look at every part in the viewer once. The viewer shows the real answer.

48

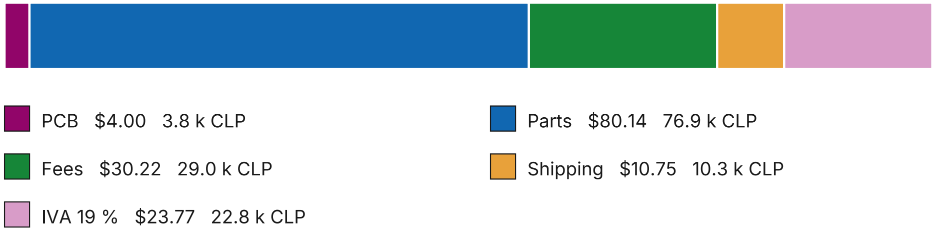
MAKING THE BOARD

What the order costs

Five assembled boards cost $114.36 at the factory. Shipping and IVA take it to $148.88, or 143 thousand CLP at 960 CLP per USD.

That is $29.78 per board. DHL Express would make the total $186.67, or 179 thousand CLP.

REAL QUOTE, 5 BOARDS, USD



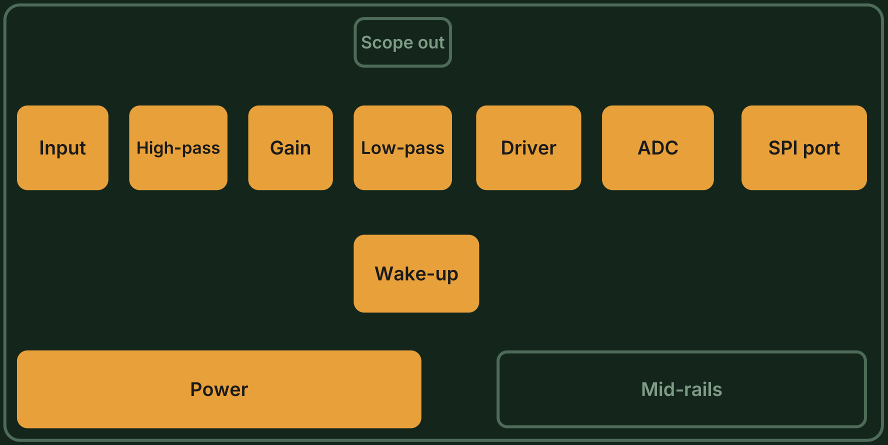
Total $148.88. 2 boards $88.48, 3 boards $108.61.

SECTION 8

Bring-up and checks

STEPS 49 TO 54

What to measure before the PMOD goes in, the first signal test, the response to expect, how to test the wake-up, what to do when something fails and what is still open.


Scope out
Input
High-pass
Gain
Low-pass
Driver
ADC
SPI port
Wake-up
Power
Mid-rails

BLOCKS AT WORK IN THIS SECTION (SEE PAGE 3)

BRING-UP AND CHECKS

49 What to check before the PMOD goes in

Feed 9 V to J2 with the PMOD unplugged. Measure each rail against GND with a meter. Unplug the supply if the 5 V rail is off by more than 0.25 V.

The 3.3 V line comes from the ZedBoard through the PMOD. With no PMOD it reads 0 V, and that is correct.

EXPECTED VOLTAGES, 9 V IN

Node	Where	Expect
+5V	C29 plus side	5.0 V
AVDD	C18 or U4 pin 6	3.30 V
VMID5	U2 pin 7	2.50 V
VOCM	R14 and R15 joint	1.65 V
+3V3	J3 pin 11	0 V (no PMOD)
nWAKE	J3 pin 8	floats, no pull-up

BRING-UP AND CHECKS

50 The first signal test

Plug the PMOD in. Put 100 mV peak at 250 kHz on J1 from the Analog Discovery 2 and send a burst of ten cycles. Look at SCOPE OUT and read the SPI words.

The scope shows 410 mV peak above 2.5 V. The ADC sees 410 mV peak between its inputs, which is 1018 codes.

EXPECTED, 100 mV PEAK AT J1

Where	Expect
J1	100 mV peak, 250 kHz
SCOPE OUT (J4)	410 mV peak on 2.5 V
ADC input, differential	410 mV peak
Code, peak	about 1018 (0x03FA)
Code, no signal	near 0 (0x0000), ± 1

BRING-UP AND CHECKS

51 What response to measure

Sweep J1 at 100 mV and read the code amplitude. Divide by the amplitude at 250 kHz. The numbers below come from the simulation.

Component spread moves the 10 kHz point. In 11 of 150 random runs the loss at 10 kHz was below 40 dB.

SIMULATION, NOMINAL PARTS

Frequency	Gain SMA to ADC	Against 250 kHz
10 kHz	-28.2 dB	-40.5 dB
50 kHz	-0.5 dB	-12.7 dB
100 kHz	+9.2 dB	-3.1 dB
250 kHz	+12.3 dB	+0.0 dB
500 kHz	+9.4 dB	-2.8 dB
1000 kHz	+0.1 dB	-12.1 dB
2000 kHz	-11.7 dB	-24.0 dB

BRING-UP AND CHECKS

52 How to test the wake-up

Sweep RV1 and keep a burst at J1. nWAKE must drop for a burst above the threshold and stay high below it. The table gives the threshold for five trimmer positions.

The lowest setting is ± 9 mV at J1. The quiet perfboard noise is 4.6 mV rms in the band, so that setting is only about 2 times the noise.

COMPUTED FROM THE RESISTORS

Trimmer	RV1	Threshold at J1
0 %	0.0 Ω	± 9 mV
25 %	5 kΩ	± 51 mV
50 %	10 kΩ	± 87 mV
75 %	15 kΩ	± 119 mV
100 %	20 kΩ	± 147 mV

ESTIMATE: comparator offset (up to 5 mV, datasheet) moves each threshold by that much.

BRING-UP AND CHECKS

53 What if it does not work

Find the first stage that is wrong, then look only at that stage. Work from the input toward the ADC. Measure DC first, then the signal.

Most first-board failures are a short, a part turned the wrong way or a missing 3.3 V.

SYMPTOM, CAUSE, CHECK

Symptom	Likely cause	Check
5 V low	Battery under 6.7 V	J2 voltage
SCOPE OUT flat	U1 or U2 not powered	5 V at pin 8
SDO all zeros	First frame only	Read twice
SDO stuck	No 3.3 V from PMOD	J3 pin 11
Wake always low	RV1 near 0 and noise	Turn RV1 up
Code offset	VOCM or VMID off	Step 49 table

NOTES

BRING-UP AND CHECKS

54 What is verified and what is open

Checked by tools: the electrical rules, the design rules, the netlist against the board, and the rotations in the JLC viewer. Checked by simulation: gain, band, noise and settling.

Nothing has run on a real board yet. Red rows mark the three open items.

STATUS

Item	Status	How
ERC, DRC, parity	Done	KiCad, 0 errors
Rotations	Done	JLC viewer
Chain response, noise	Simulated	ngspice, vendor models
THS4531, 400 pF load	OPEN	LTspice netlist, not run
PMOD fit on ZedBoard	OPEN	print 1:1 and test
First board, bench	OPEN	steps 49 to 52

NOTES

WHERE THINGS LIVE

Files, models and datasheets

`hw/`	KiCad project, schematic and board
`hw/fab/`	Gerbers, BOM and CPL for JLCPCB
`hw/tools/`	Generators, DRC, cost model, 1:1 print
`sim/`	Chain simulation, vendor models list, LTspice netlist
`doc/tools/`	This manual, built from the files above
`doc/data/`	Board and simulation snapshots used in the figures

DATASHEETS

ADS7057, THS4531, OPA365 (OPA2365), LM393, XC6206, AMS1117

```
Rebuild: M4_PDF_PYTHON=<python with pymupdf> python3 doc/tools/make_adc_manual.py
```